

Trainings ▾

General Information

C Command HD Elite™ Marine Panel System

- The C Command HD Elite™ Marine panel system is used to monitor engine operating characteristics and control local and remote start and stop functions.
- This system receives engine data information and has the ability to receive Society of Automotive Engineers (SAE) J1939 data link information from the engine control module (ECM), and has connections for additional vessel sensors. The C Command HD Elite™ Marine panel system is comprised of the following panels:
 - Customer Interface Box (C.I.B.) with diesel control unit
 - Remote panel.

Customer Interface Box

The C.I.B. is located in the engine room and receives engine data from the ECM through one 47 pin connector and one 31 pin connector. A 9 pin service connector is also provided for electronic service tool connection. The engine receives start and stop commands through the C.I.B. The C.I.B. contains the following components:

- Diesel Control Unit - Provides engine control and monitoring
- Shutdown Unit - Provides engine protection via shutdown if critical parameters are exceeded
- Crank Relay - Provides logic function for engine cranking
- Energize to Stop Relay - Provides logic function for engine stop sequence
- Circuit breakers - Prevents system overcurrent draw
- Terminal strips - Provides connection points for wires
- Trim Potentiometers - Allows for Gain (QSM11 **only**), Droop, and Frequency adjustment
- Engine stop button - Allows direct engine stop from the C.I.B.
- Redundant Power Selector - Provides a constant power supply to the diesel control unit.

Engine Protection Override Relay- C Command HD Elite™ Marine panel system PS106

- The C Command Elite™ panel system is equipped with the ability to override an engine protection shutdown command from the ECM. If the engine protection override function is desired, the original equipment manufacturer (OEM) is responsible for the installation of the switch.
- The engine protection override terminal is located on the X4 terminal strip in the C.I.B.
- When the engine protection override switch is activated, a signal is sent to the ECM through the 31 pin connector. This allows the ECM to disregard the engine protection shutdown.
- This relay also overrides diesel control unit shutdowns, if enabled.

Engine Protection Override Relay - C Command HD Elite™ Marine panel system PS107

- The C Command Elite panel system is equipped with the ability to override an engine protection shutdown command from the ECM and DCU. If the engine protection override function is desired, the original equipment manufacturer (OEM) is responsible for the installation of the switch.
- The engine protection override terminal is located on the X1 terminal strip in the C.I.B.
- When the engine protection override switch is activated, a signal is sent to the ECM through the 47 pin connector. This allows the ECM to disregard the engine protection shutdown.
- This relay also overrides diesel control unit shutdowns, if enabled.
- The diesel control unit allows to configure the engine protection override switch signal transmitted to Shutdown Unit. This allows disregarding engine protection shutdown triggered by ECM, diesel control unit, shutdown unit if enabled.
- The diesel control unit engine protection override switch signal configurable setting with the use of web-based configuration interface.

[AG1]Last edit

Diesel Control Unit

The diesel control unit 410(E) is an electronic unit that is able to read engine data. The data can be supplied from a SAE J1939 communication data bus, and from switches and sensors. The data is displayed on a liquid crystal display.

The diesel control unit 410(E) contains the following:

- Adjustable set points for warning, alarm, and shutdown
- An alarm list that displays an overview for all active alarms
- A diagnostic list that displays active diagnostic messages
- Ethernet and Modicon™ communication bus channels available for communication to third parties.

It is possible to update the diesel control unit firmware using a laptop. The diesel control unit 410(E) operates on 24 VDC and can tolerate 24 VDC (connected in error) on all input/output points, regardless of type.

The diesel control unit 410(E) requires two power supplies. The main purpose for two power supplies is to make sure sufficient supply voltage is present upon cranking the engine. If a secondary supply was **not** available and the primary supply voltage drop was significant, it can possibly cause the reboot of the diesel control unit 410(E).

Some C.I.B. may use a redundant power selector instead of a power switch relay. The redundant power selector is a power distributor for both the primary and secondary power supply. Its purpose is to effortlessly switch between the primary and secondary power supply to provide a constant output voltage. It can communicate with the vessel operator to notify when the power source has changed between the two sources.

Liquid Crystal Display

- 320 x 240 pixel color graphic display for 410

- 680 x 480 pixel color graphic display for 410(E)
- Backlight and front panel LEDs adjust automatically with the ambient light conditions, using a light sensor in the front panel
- Backlight can be set to 10 percent, 50 percent, or full intensity from the menu
- Any button or any event (start, stop, alarm, etc) will turn on the backlight.

Backlight Intensity	
Ambient Light Intensity	Backlight
Completely dark	10 percent
Direct sun	100 percent

Light Emitting Diode

- Alarms
 - Dual color; red and amber
 - Amber will be lit on warning
 - Red will be lit on alarm
 - If both are active, red has priority over amber. Red and amber will **not** be lit simultaneously
 - If acknowledged red and new amber, flash amber.
- Start button
 - Green lit, solid steady when engine is running
 - 2 Hz flashing when engine is cranking
- Stop button
 - Red lit, solid steady when engine is stopped
 - 1 Hz flashing when engine is stopping, if caused by a normal stop (stop button, remote stop, etc.)
 - 2 Hz flashing when engine is stopping due to a sensor configured to make the engine shut down (overspeed, etc.)
- Power light emitting diode
 - Green lit, sold when powered
 - Immediately 1 Hz flashing when voltage is below set point, or one of the two supplies is missing.

Alarms/Light Emitting Diode Functional Summary	
Amber	Red
Amber ECM diagnostic fault	Red ECM diagnostic fault
Warning diesel control unit	Alarm diesel control unit 410 panel fault
	Shutdown diesel control unit 410 panel fault

The diesel control unit 410(E) has a 24 VDC sleep mode input. This allows the unit to be set into a mode that resembles a unit that is turned off.

- 24 VDC = system on
- 0 VDC = sleep mode

Sleep mode is controlled by the power on/off switch on the front of the C.I.B.

Local, Remote, Automatic and Manual Functions

- Local mode
 - No external commands work (wire terminal commands or communication commands)
 - **Only** local panel buttons work.
- Remote mode
 - Local and remote commands work equally.
- Automatic mode
 - Automatic start wire inputs work
 - Automatic stop wire inputs work.
- Manual mode
 - Automatic start wire inputs will **not** work
 - Automatic stop wire inputs will **not** work.

When changing between modes, a dialog will appear with the description as shown in the table below:

Dialog Boxes When Changing Modes	
When changing to:	Dialog box
Local mode	External commands are disabled
Remote mode	External commands are enabled
Automatic mode	Automatic start/stop is enabled
Manual mode	Automatic start/stop is disabled

Diesel Control Unit Buttons and Functions

- Start button
 - Press to start engine
 - Cranking output relay is activated
 - Diesel control unit 410(E) will continue its start attempt until engine has started, finished the start attempt, or the user has aborted the start by pressing the stop button.
- Prelubrication
 - Diesel control unit has engine prelubrication control capabilities
 - Prelubrication cycle = prelubrication activation signal - await completion - crank
 - If the diesel control unit 410(E) does **not** receive activation signal within 30 seconds, "Start Failed, Prelube Fault" dialog is generated on the display
 - While prelubrication is in progress "Prelubrication in progress" dialog is generated on the display.
- Automatic starts
 - The diesel control unit 410(E) will initiate as many start attempts according to the configured number of start attempts in the configuration menu.
- Stop button
 - Press to stop a running engine or to abort a start sequence
 - Depending on the configuration, the button **must** be pushed until the engine is stopped
 - A stop sequence can be aborted by pressing the start button
 - While in stopping mode, "Stop in progress - press Start to cancel" dialog is generated on the display.

- Alarm list button
 - When pressed, displays the active alarm list and silences the panel buzzer.
- Up arrow button
 - When pressed, moves selection up a selectable list.
- Down arrow button
 - When pressed, moves selection down a selectable list.
- Menu/Home button
 - When pressed, opens the diesel control unit 410(E) menu.
 - All functions described in this document are accessible from the menu.
 - The menu also includes diesel control unit 410(E) configuration and test options.
- Buzzer and buzzer off
 - Buzzer activates when new alarms or diagnostic messages are received by the diesel control unit 410(E)
 - Press the alarm list button to silence the buzzer
 - If a new alarm is received while in alarm list, the buzzer will activate. Pushing the alarm list button will silence the buzzer.
- Soft buttons
 - Change functionality depending on the actual screen
 - Four buttons located along the bottom of the display
 - Dynamic label is linked to each button.

Soft Button Labels in Main View

Backlight	Local Start Only/All Start	Settings	User Configuration
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Prelubrication Operation

- Prelubrication is necessary on engines that need to build up oil pressure prior to cranking.
- If prelubrication is activated, the diesel control unit will begin a sequence when the start button is pushed.
 - Receives start command (local, remote, or automatic)
 - Prelubrication is activated for a fixed time or until the oil pressure is acceptable
 - After fixed time has elapsed or oil pressure is acceptable, engine will crank
 - If pressure is timed out, crank to start failure or prelubrication override.

Alarm Page

The alarms appear in the order they are received. When a new alarm is received, it is shown at the top of the alarm list.

The list shows all alarms from configured switches and analog inputs. These can be from physical switches, 4-20 mA type inputs, and configured setpoints on the SAE J1939 data link (ECM broadcasted) channels.

The symbol numbers and colors denote the severity of the alarm:

- Symbol 1 = Red severity
- Symbol 2 = Amber severity

- Symbol 3 = White severity.

Bold face lettering denotes alarm acknowledgement:

- Bold = Unacknowledged
- Normal = Acknowledged.

Reverse background and foreground color denotes active status:

- Colored background = Active alarm
- Colored text on gray background = Inactive alarm.

Panel Alarms

- Located on the on the alarm page, displays alarms from diesel control unit, shutdown unit and remote input/output unit(shutdown unit and remote input/output are Marine C Command Elite™ components **only**).
- Red = shutdown (switch closure on shutdown unit)
- Amber = alarm (alarm sensor threshold to remote input/output unit exceeded or wire breaks for shutdown unit switch inputs)
- White = warning (out of range remote input/output malfunctions on remote input/output unit sensor inputs).

Event Log

- Event log stores in a separate list the last events. An event is defined as:
 - Any new panel alarm that appears in the alarm list
 - Any new engine diagnostic message that appears in the alarm list
 - When alarm or diagnostic messages go to inactive
 - When any user interaction buttons are pressed, except for up and down arrows
 - Any command such as remote start, stop, acknowledge, etc.

The event log can contain 500 separate events. When the log is full and a new event is logged, the oldest event is removed.

The event will be time stamped.

Service Intervals

- There are four different timers with corresponding text for service intervals. All are set to off by default.
- Intervals are enabled during initial configuration. Enabling the interval starts the timer.
- When the time until next service is 10 percent of the total interval hours, a dialog appears when entering ready mode, displaying “Next Service Due” in (10 percent of total interval hours).
- If a service is overdue, a dialog box appears displaying “Service Overdue” (amount of hours overdue).
- All service interval actions are in the event log.

Communication Interfaces

- COM1 - Modicon Communication Bus Parent (shutdown unit link)
 - Immediately starts communication when shutdown unit is corrected. **Only** used in Marine C Command Elite™ panel system.

- COM2 - Communication Bus Parent (input/output expansion to remote input/output unit 410(E))
 - Immediately starts communication when expansion input/output expansion is connected. **Only** used in Marine C Command Elite™ panel system.
- COM3 - Modicon Communication Bus Child (configuration and general communication)
 - Communication with third party equipment to communicate with the diesel control unit.
- COM4 - J1939/Controller Area Network (ECM SAE J1939 interface)
 - Connected to the engine ECM.
- COM5 - Controller Area Network (ECM SAE J1939 interface)
 - Three wire terminals (high, low, and shield)
- COM6 - Ethernet Modicon Communication Bus/Transmission Control Protocol (general communication and web server configuration)
 - Primarily for Ethernet Modicon Communication Bus/Transmission Control Protocol, but can be used for limited configuration of diesel control unit
- COM7 - Console
 - Universal serial bus connector for processor interface.

The Marine C Command Elite™ panel system has the following inputs supplied from the ECM:

Propulsion

- Torque curve select signal
- Droop select signal
- SAE J1939 supply signal
- SAE J1939 return signal
- External speed command input signal
- OEM switch signal
- Intermediate speed control 3/validate switch signal
- Intermediate speed control 2 switch signal
- Engine protection override switch signal
- Diagnostic test mode switch signal
- Alternate idle/auxiliary governor select signal
- Intermediate speed control/idle increment signal
- Intermediate speed control/idle decrement signal
- Intermediate speed control 1/remote power takeoff (PTO) signal
- Fan control switch signal
- Tachometer output signal
- Frequency accelerator signal
- Accelerator pedal or lever idle validation switch off signal
- Accelerator pedal or lever idle validation switch on signal
- Maintenance signal
- Warning signal
- Engine protection stop signal
- Fan control signal
- Multiple unit synchronization switch 2 signal
- Multiple unit synchronization switch 1 signal

- Remote accelerator pedal or lever position switch signal
- Engine oil level remote reservoir signal
- Remote accelerator pedal or lever signal
- Auxiliary input/output 3 signal
- Auxiliary PWM driver signal
- Multiple unit synchronization pin 3 switch signal
- Multiple unit synchronization pin 2 switch signal
- Multiple unit synchronization pin 1 switch signal.

Auxiliary

- SAE J1939 data link supply signal
- SAE J1939 data link return signal
- Fault acknowledge signal
- Idle/rated switch signal
- Engine protection override switch signal
- Diagnostic test mode switch signal
- Diagnostic increment signal
- Diagnostic decrement signal
- Alternate frequency signal
- Coolant temperature meter signal
- Tachometer output signal
- Generator speed/load governing bias signal
- Low oil pressure shutdown signal
- Run/stop switch signal
- Overspeed switch shutdown signal
- Common warning signal
- Common shutdown signal
- Engine oil remote reservoir signal
- Droop adjust potentiometer signal
- Frequency adjust signal
- Oil pressure meter signal
- Multiple unit synchronization id pin 3 switch signal
- Multiple unit synchronization id pin 2 switch signal
- Multiple unit synchronization id pin 1 switch signal
- Pre-high engine temperature warning signal
- High engine temperature shutdown signal
- Pre-low oil pressure warning signal.

DC410

- The diesel control unit 410 has an input/output as a standard. The following inputs and outputs are available on the unit.

Standard Input/Output Arrangement for Diesel Control Unit 410			
Type	Input or Output	Number of Channels	Description
Fixed Function Inputs	Input	10	Fixed Input

Standard Input/Output Arrangement for Diesel Control Unit 410

Type	Input or Output	Number of Channels	Description
Configurable Fixed Functional Inputs	Input	2	None
Switch	Input	8	User Configurable
4-20 mA	Input	4	User Configurable
PT100	Input	4	User Configurable
Fixed 24 VDC Source 500 mA	Output	12	Fixed
Configurable Outputs 24VDC 500 mA	Output	4	User Configurable (multiple relay)
Configurable outputs, relays	Output	2	User Configurable (multiple relay)
Pickup (Tachometer)	Input	1	User Configurable
SAE J1939	Input	1	Fixed/User Configurable
SAE J1939 Open	Output	1	None
Ethernet (Modicon Communication Bus/ Transmission Control Protocol)	Input/Output	1	None

The tables below are a breakdown of the type of input/output for the diesel control unit 410.

Fixed Functional Inputs (Active on 24 VDC)

Channel Number	Function	Active	Description
1	Remote acknowledge	24 VDC = acknowledge	Remote acknowledge and silence alarm.
2	Automatic start (power management system)	24 VDC = start	Initiate automatic start attempts. Works in manual mode only .
3	Delayed automatic stop (power management system stop)	24 VDC = stop	Works in automatic mode only . If power management system start and stop are active, power management system start is given priority.
4	Remote start	24 VDC = start	None
5	Remote stop	24 VDC = stop	None

Fixed Functional Inputs (Active on 24 VDC)			
Channel Number	Function	Active	Description
6	Start disabled	24 VDC = start disabled	Fail to safe = able to start. Does not stop a running engine; prevents a new start.
7	Prelubrication complete	24 VDC = complete	Must have 24 VDC to start the engine.
8	Gearbox neutral	24 VDC = in gear	Fail to safe = will start even if wire breaks. Gearbox is disabled with oil pressure alarm. Engine crank is disabled when in gear.
9	Manual/Automatic Automatic = power management system control. Manual = power management system control lockout.	24 VDC = manual Open = automatic	Fail to safe = automatic. If manual is selected in the diesel control unit 410, remote panel, or at wire terminal, command is manual.
10	Shutdown override	None	None
Configurable Fixed Functional Inputs			
Channel Number	Function	Description	
1	Configurable	Configurable from a bank of fixed functions.	
2	Configurable	Configurable from a bank of fixed functions.	
Configurable Switch Inputs (24 VDC Inputs, Warning, or Alarm)			
Channel Number	Function	Description	
1	Open	Normally closed/normally open	
2	Open	Normally closed/normally open	
3	Open	Normally closed/normally open	
4	Open	Normally closed/normally open	

Configurable Switch Inputs (24 VDC Inputs, Warning, or Alarm)		
Channel Number	Function	Description
5	Open	Normally closed/normally open
6	Open	Normally closed/normally open
7	Open	Normally closed/normally open
4-20 mA Inputs (Two Wire Terminals per Channel)		
Channel Number	Range	
1	Configurable	
2	Configurable	
3	Configurable	
4	Configurable	
PT100 Inputs (Three Wire Terminals per Channel)		
Channel Number	Range	
1	Configurable	
2	Configurable	
3	Configurable	
4	Configurable	
Fixed Functional Outputs (24 VDC Source 500 mA)		
Channel Number	Function	Description
1	Prelubrication activation	When configured, precedes engine crank.
2	Crank (start)	None
3	Engine running	None
4	Energize to run	None
5	Energize to stop	Also used on shutdown stop in applications that are classified.
6	Energize to shut down	Activates for shutdowns only .
7	Configurable	Any function.
8	Configurable	Any function.
9	Configurable	Any function.
10	Common alarm relay (three terminals)	Inclusive common warning and common diagnostics. 24 VDC = Normal (no alarm). Normally energized.

Fixed Functional Outputs (24 VDC Source 500 mA)		
Channel Number	Function	Description
11	Common alarm relay	Normally open.
12	Common alarm	Normally closed.
Configurable Function Outputs		
Channel Number	Description	
1	User configurable (multiple relay)	
2	User configurable (multiple relay)	
3	User configurable (multiple relay)	
4	User configurable (multiple relay)	
Configurable Function Outputs - Relay		
Channel Number	Description	
1	User configurable (multiple relay)	
2	User configurable (multiple relay)	

Engine Speed Input

- The diesel control unit 410 will read up to three sources (speed input, pickup, and tachometer) and use one for calculation and display as follows:
 - First priority - SAE J1939 from the ECM
 - Second priority - Physical input in shutdown unit, via communication link
 - Third priority - Physical input on diesel control unit 410 (pickup/tachometer).

The following tables have specifications for the functional inputs/outputs and sensor inputs available for the diesel control unit 410.

Functional Inputs	
Input	Description
Fixed functional inputs	Optocoupled switch input. 0 to 2 VDC = 0 7 to 32 VDC = 1 2 to 7 VDC = undefined Over voltage protection: 40 VDC Not connected = 0 VDC (used with 47k ohm resistor)

Functional Inputs	
Input	Description
General switch inputs	Optocoupled switch input. 0 to 2 VDC = 0 7 to 32 VDC = 1 2 to 8 VDC = undefined Over voltage protection: 40 VDC Not connected = 0 VDC (used with 47k ohm resistor) To be configured for: Warning/alarms/shutdowns Delay for warnings/alarms/shutdowns Normally open or Normally closed On run only .
Sensor Inputs	
Sensor	Description
4 to 20 mA sensor	Analog input 0 to 25 mA, with a guaranteed specification for 4 to 20 mA measurements Accuracy: 0.5 percent over entire range. Over voltage protection 40 VDC Not connected = 0 mA Alarm for: Less than 2 mA Greater than 25 mA
PT100 sensor	3-wire channels Range 0 to 800° C [0 to 1472°F] Accuracy: 0.5 percent over entire range
Pickup/Tachometer/Speed Input	0 to 10 kHz
SAE J1939	60 VDC transceiver Wires: Supply, Return, and Shield
Functional Outputs	
Output	Description
24 VDC outputs, 500 mA	Not active = Open (47k ohm resistor to define voltage level) Active = 24 VDC to 0 VDC Polyswitch over current protection
DCU 410E The diesel control unit 410(E) has an input/output as a standard. The following inputs and outputs are available on the unit.	

Standard Input/Output Arrangement for Diesel Control Unit 410			
Type	Input or Output	Number of Channels	Description
Fixed Function Inputs	Input	10	Fixed Input
Configurable Fixed Functional Inputs	Input	2	None
Switch	Input	8	User Configurable
4-20 mA	Input	4	User Configurable
PT100	Input	4	User Configurable
Fixed 24 VDC Source 500 mA	Output	5	Fixed
Fixed 5 VDC Source 500 mA	Output	1	Fixed
Configurable Outputs 24VDC 500 mA	Output	7	User Configurable (multiple relay)
Configurable outputs, relays	Output	2	User Configurable (multiple relay)
Pickup (Tachometer)	Input	1	User Configurable
SAE J1939	Input	2	Fixed/User Configurable
Ethernet (Modicon Communication Bus/ Transmission Control Protocol)	Input/Output	1	None
USB	Input/Output	1	None
The tables below are a breakdown of the type of input/output for the diesel control unit 410(E).			
Fixed Functional Inputs (Active on 24 VDC)			
Channel Number	Function	Active	Description
1	Remote acknowledge	24 VDC = acknowledge	Remote acknowledge and silence alarm.
2	Automatic start (power management system)	24 VDC = start	Initiate automatic start attempts. Works in manual mode only .
3	Delayed automatic stop (power management system stop)	24 VDC = stop	Works in automatic mode only . If power management system start and stop are active, power management system start is given priority.

Fixed Functional Inputs (Active on 24 VDC)			
Channel Number	Function	Active	Description
4	Remote start	24 VDC = start	None
5	Remote stop	24 VDC = stop	None
6	Start disabled	24 VDC = start disabled	Fail to safe = able to start. Does not stop a running engine; prevents a new start.
7	Prelubrication complete	24 VDC = complete	Must have 24 VDC to start the engine.
8	Gearbox neutral	24 VDC = in gear	Fail to safe = will start even if wire breaks. Gearbox is disabled with oil pressure alarm. Engine crank is disabled when in gear.
9	Manual/Automatic: Automatic = power management system control. Manual = power management system control lockout.	24 VDC = manual Open = automatic	Fail to safe = automatic. If manual is selected in the diesel control unit 410(E), remote panel, or at wire terminal, command is manual.
10	Shutdown override	None	None
Configurable Fixed Functional Inputs			
Channel Number	Function	Description	
1	Configurable	Configurable from a bank of fixed functions.	
2	Configurable	Configurable from a bank of fixed functions.	
Configurable Switch Inputs (24 VDC Inputs, Warning, or Alarm)			
Channel Number	Function	Description	
1	Open	24 VDC Switch Input	
2	Open	24 VDC Switch Input	
3	Open	24 VDC Switch Input	
4	Open	24 VDC Switch Input	
5	Open	24 VDC Switch Input	
6	Open	24 VDC Switch Input	

Configurable Switch Inputs (24 VDC Inputs, Warning, or Alarm)		
Channel Number	Function	Description
7	Open	24 VDC Switch Input
8	Open	24 VDC Switch Input
4-20 mA Inputs (Two Wire Terminals per Channel)		
Channel Number	Range	
1	Configurable	
2	Configurable	
3	Configurable	
4	Configurable	
PT100 Inputs (Three Wire Terminals per Channel)		
Channel Number	Range	
1	Configurable	
2	Configurable	
3	Configurable	
4	Configurable	
Fixed Functional Outputs (24 VDC Source 500 mA)		
Channel Number	Function	Description
1	Prelubrication activation	When configured, precedes engine crank.
2	Crank (start)	None
3	Engine running	None
4	Energize to run	None
5	Energize to stop	Also used on shutdown stop in applications that are classified.
6	Energize to shut down	Activates for shutdowns only .
7	Common alarm relay (three terminals)	Inclusive common warning and common diagnostics. 24 VDC = Normal (no alarm). Normally energized.
8	Common alarm relay	Normally open.
9	Common alarm	Normally closed.
Configurable Function Outputs		
Channel Number	Description	
1	User configurable (multiple relay)	
2	User configurable (multiple relay)	

Configurable Function Outputs	
Channel Number	Description
3	User configurable (multiple relay)
4	User configurable (multiple relay)
5	User configurable (multiple relay)
6	User configurable (multiple relay)
7	User configurable (multiple relay)
Configurable Function Outputs - Relay	
Channel Number	Description
1	User configurable (multiple relay)
2	User configurable (multiple relay)
Engine Speed Input - The diesel control unit 410(E) will read up to three sources (speed input, pickup, and tachometer) and use one for calculation and display as follows: - First priority - SAE J1939 from the ECM - Second priority - Physical input in shutdown unit, via communication link - Third priority - Physical input on diesel control unit 410(E) (pickup/tachometer). The following tables have specifications for the functional inputs/outputs and sensor inputs available for the diesel control unit 410(E).	
Functional Inputs	
Input	Description
Fixed functional inputs	Optocoupled switch input. 0 to 2 VDC = 0 7 to 32 VDC = 1 2 to 7 VDC = undefined Over voltage protection: 40 VDC Not connected = 0 VDC (used with 47k ohm resistor)
General switch inputs	Optocoupled switch input. 0 to 2 VDC = 0 7 to 32 VDC = 1 2 to 8 VDC = undefined Over voltage protection: 40 VDC Not connected = 0 VDC (used with 47k ohm resistor) To be configured for: Warning/alarms/shutdowns Delay for warnings/alarms/shutdowns Normally open or Normally closed On run only .

Sensor Inputs	
Sensor	Description
4 to 20 mA sensor	Analog input: 0 to 25 mA, with a guaranteed specification for 4 to 20 mA measurements Accuracy: 0.5 percent over entire range. Over voltage protection 40 VDC Not connected = 0 mA Alarm for: Less than 2 mA Greater than 25 mA
PT100 sensor	3-wire channels Range 0 to 800°C [0 to 1472°F] Accuracy: 0.5 percent over entire range
Pickup/Tachometer/Speed Input	0 to 10 kHz
SAE J1939	60 VDC transceiver Wires: Supply, Return, and Shield
Functional Outputs	
Output	Description
24 VDC outputs, 500 mA	Not active = Open (47k ohm resistor to define voltage level) Active = 24 VDC to 0 VDC Polyswitch over current protection
Remote Panel - Connected to the diesel control unit 410(E), and is remote means of controlling the diesel control unit 410(E). - When connected to the diesel control unit, remote panel will update itself with diesel control unit 410(E) configuration files. - Touch screen control	
Power Light Emitting Diode - Illuminated when supply power is OK. - Flashing 1 Hz when power is below fixed set point. - Ambient light sensor located in top right area of panel. - Ambient light sensor is for display back light.	
Display Main Screens - There are four different main screens, depending on the number of diesel control units (engines) that are connected to the engine. - The screen is viewable in direct sunlight (8.4-inch diagonal, 640 x 480 pixel transfective screen) - The remote panel will display as follows:	

- Type 1 - One diesel control unit is connected to the remote panel. The entire display screen is used.
- Type 2 - Two diesel control units are connected to the remote panel. The display screen is divided into two quadrants.
- Type 4 - Three or four diesel control units are connected to the remote panel. The display screen is divided into four quadrants.
- Type 6 - Five or six diesel control units are connected to the remote panel. The display screen is divided into six quadrants.

Communication

- Ethernet Modicon Communication Bus/Transmission control protocol for communication to the diesel control units.
- The remote panel will give an alarm if communication is lost with diesel control unit 410(E) or other remote panels. RS232 for communication to optional external vessel inputs.

Functionality

- Main page
 - Start up page and will find all diesel control units and list them by engine name.
- Engine name
 - Part of configuration of diesel control unit. Each configured diesel control unit will have a unique name. If two or more names are the same in diesel control unit configuration, the remote unit will add a numeric value to the end of the names
- Ignore engines
 - On each remote panel, it will be possible to ignore one or several diesel control unit panels in the network. A diesel control unit that is ignored at the remote unit will be "invisible" for that remote unit.
- Active station
 - **Only** one remote panel will have control when more than one remote panel is in the network. An active station can execute commands such as: start, stop, and alarm acknowledge.
- Requesting Control
 - When an inactive remote panel requests to be active, the active station receives a dialog box for hand over.
- Remote priorities

In the configuration menu, the remote unit is given a priority number as follows:

Remote Panel Priorities			
Priority 1 (Primary Parent Remote Panel)	Priority 2 (Secondary Parent Remote Panel)	Priority 3 (Child Remote Panel) [Factory default setting]	Priority 4 (Public Remote Panel)
Can always take command from the active panel; no question asked. Note: If the user tries to execute a command, and the panel is not the active station, it immediately becomes the active station and executes the command	Can take command from active panels priority 2 or 3; no question asked. If the current active panel is priority 1, that panel must confirm.	Can take command from active panels priority 1, 2, or 3 after confirmation at the active panel.	Disabled
Will hand over command to priority 1 panel immediately. Can hand over command to priority 2 or 3 panels, depending upon Confirmation Time-out Behavior	Will hand over command to priority 1 or 2 panels immediately. Can hand over command to priority 3 panels, depending upon Confirmation Time-out Behavior.	Will hand over command to priority 1 or 2 panels immediately. Can hand over command to priority 3 panels, depending upon Confirmation Time-out Behavior.	Will hand over command to priority 1, 2, or 3 panels immediately.

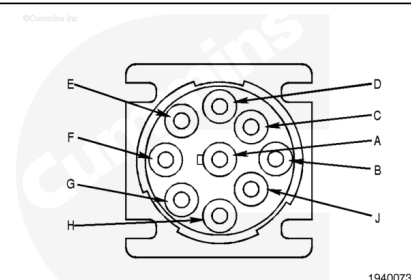
If the present active station suddenly loses communication in the network, the first panel that asks for control will immediately become the active station.

Configuration

- The remote panel has a limited configuration requirement
- The remote panel will find diesel control units available on the Ethernet network and update its configuration files.

Datalink Connectors

Cummins Inc. produces many engines today that are electronically controlled. These engines have special diagnostic requirements because of the ECM in the system. To interface with these ECMs, electronic service tools have been developed, such as the recommended Cummins® electronic service tool. The Cummins® electronic service tool interfaces with the electronic engines by means of a data link. A data link



provides a physical means for transmitting and sorting electronic signals. A data link consists of special electronic circuitry and electrical harnesses. Connection points for electronic service tools are also part of the data link. Data links are defined by standards written by the SAE. Cummins Inc. uses two such standards for electronic service tools. One is a combination of SAE J1587/SAE J1708 and the other is SAE J1939. Engines can support one or both of these data link standards

The recommended data link connector for Cummins® engines is a 9 pin Deutsch™ connector. This connector can supply SAE J1587/SAE J1708 and SAE J1939 communications and battery voltage. The following are pin-outs for the 9 pin connector:

Pin	Signal
A	Battery Voltage 1 Return
B	Battery Voltage 1 Supply
C	SAE J1939 Data Link Supply
D	SAE J1939 Data Link Return
E	SAE J1939 Data Link Shield
F	Not Used
G	Not Used
H	Not Used
J	Not Used